

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Problem Statement & Background

Problem Statement:

When one GaN transistor switches, its fast voltage swing pushes charge into the gate of the other one - the one that must stay off. That gate rises to **1.65 V**, and **1.4 V is enough to turn it on**. Both conduct, and the supply shorts through them.

Background & Significance:

GaN half-bridges sit inside EV inverters and chargers. GaN is chosen because it switches fast - and that speed is what causes this fault.

Existing Solutions:

Drive in steps, clamp the off gate, adjust the dead time, hold the gate at -2 V [1]-[4]. Reported: 30.5 % less overshoot, 75 % less turn-off loss [5]-[7].

Limitations / Research Gap:

Those gains are reported as one number. They mix picking one good setting with changing it while the converter runs. Only the second needs a sensor, a lookup table and a controller - so nobody knows whether that hardware is worth building.

Aim and Approach

Aim:

How much benefit does runtime adaptation add beyond a fixed gate-driver setting, and under what conditions is that benefit worth the hardware?

Proposed Solution:

A GaN buck converter driven by a **segmented gate driver**: 8 pull-up steps, 8 pull-down, adjustable dead time, a Miller clamp and a -2 V off rail - every one a setting we can change and measure.

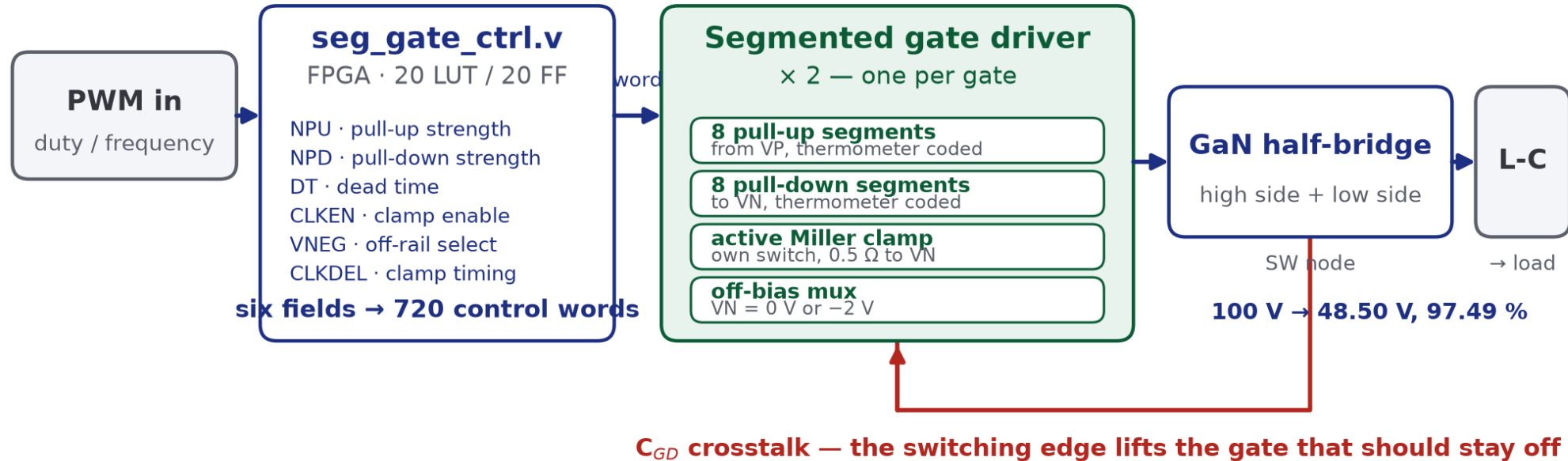
How we approached it:

- 1. Build the converter and check it converts.** 100 V in, 48.50 V out at 4.869 A: 236.26 W, 97.49 % efficient.
- 2. Recreate the fault.** At the fastest setting the off gate reaches 1.65 V against a 1.4 V threshold.
- 3. Fix it one change at a time,** each measured on its own run.
- 4. Ask whether one fixed setting is enough,** by seeing whether the best setting moves with the operating point.

Tools:

ngspice runs every simulation. **KiCad** draws every schematic, from the same netlist ngspice runs.

System Architecture



The control word is set once at power-up — no sensor, no lookup table. Re-tuning it while the converter runs is worth a further 2.6 %.

Fig. 1 - Drawn diagram, not simulation output. One signal path: a PWM command, the FPGA controller that holds the control word, the segmented driver that shapes the edge, and the half-bridge it drives. The driver in green is what this project designs; it is drawn once as [kicad/gan_segdrv.kicad_sch](#) and placed twice in [kicad/gan_buck.kicad_sch](#), one per gate. The control word is set at power-up and never changed - re-tuning it while the converter runs is the 2.6 % this project measures. In red, the fault: the switching edge couples through C_{GD} into the gate that should stay off.

Novelty: Three Added Blocks

Same command, same segmented output stage, same power stage. Three blocks added - in green.

	BASE PAPER — Zhang et al., ISPSD 2020	OURS
Control of the driver	one analogue bias resistor, set at fabrication	seg_gate_ctrl.v — digital, re-writable while it runs
Output stage	7 + 7 segments, two stages	8 + 8 segments, thermometer coded
Active Miller clamp	none	always on
Gate off rail	0 V	selectable 0 V / -2 V
Crosstalk margin at 100 V / 10 A	+0.407 V	+2.576 V

Shaded: the three blocks we add. Everything else is theirs, unchanged — same command, same power stage. 6.3× the margin, from one fixed setting.

Fig. 2 - Drawn diagram, not simulation output. The five things that differ, and nothing else. Their column is the base paper as we rebuilt it in [models/zhangdrv.lib](#); ours is [models/segdrv.lib](#). Shaded: the three we add - digital control in place of one fixed resistor, an always-on Miller clamp, and an off rail selectable to -2 V. The command, the power stage and the idea of segmenting the output are all our implementation of theirs. The last row is what those three deliver at the headline corner, read from [results/headtohead.csv](#): +0.407 V becomes +2.576 V, from one fixed setting.

Crosstalk Mechanism at Device Level

Why a fast edge turns on the device that should be off, and what the driver does about it.

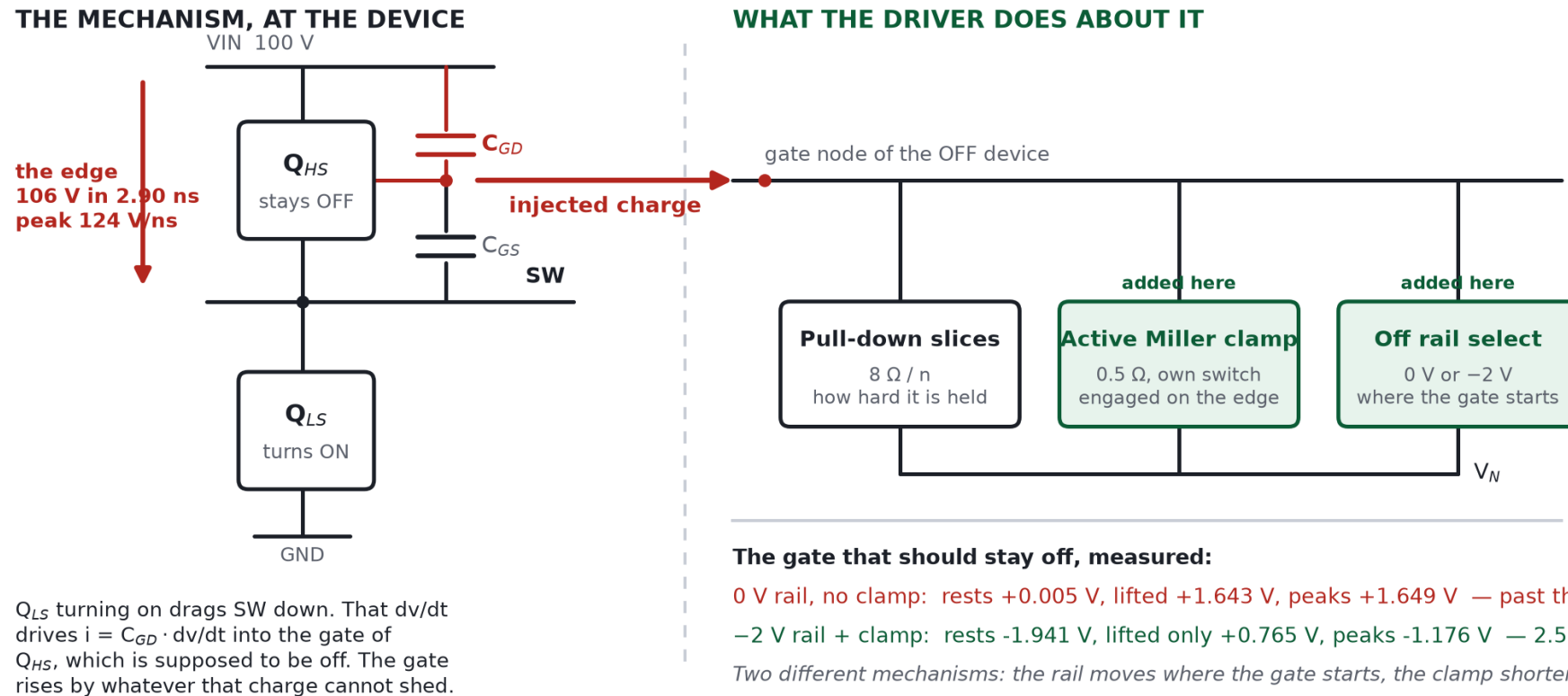


Fig. 4 - Left: the half-bridge. When the low-side device turns on the switch node falls from 106 V, its 90-10 % edge taking 2.90 ns and peaking at 124 V/ns, and that dv/dt drives a current through C_{GD} into the high-side gate. Right: the three parallel paths that current sees at the gate node. The segment count sets how hard the gate is held; the Miller clamp adds a 0.5 ohm path of its own, timed to the other device's edge; the off rail sets where the gate starts from. The two additions are not one mechanism - the rail moves the starting point, the clamp shortens the lift - which is why they add rather than overlap. [scripts/gan_level_figure.py](#), from [results/waveform_anatomy.txt](#).

Implementation: Converter Schematic

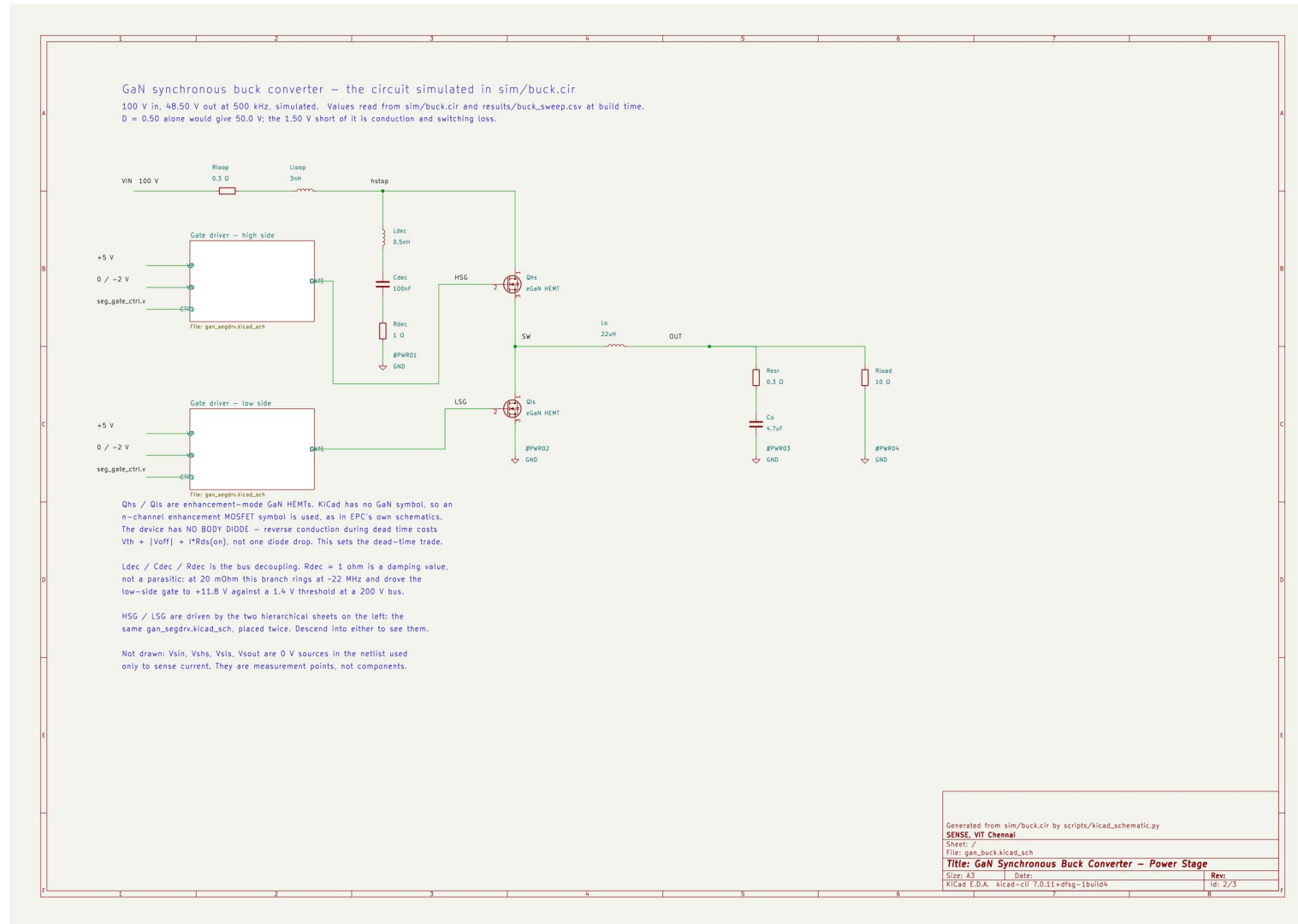


Fig. 5 - `kicad/gan_buck.kicad_sch`, the root sheet, plotted by KiCad 7.0.11. Drawn from [sim/buck.cir](#), so every value on it is the netlist's. The two gate drivers are hierarchical sub-sheets of this page.

Implementation: Proposed Gate Driver

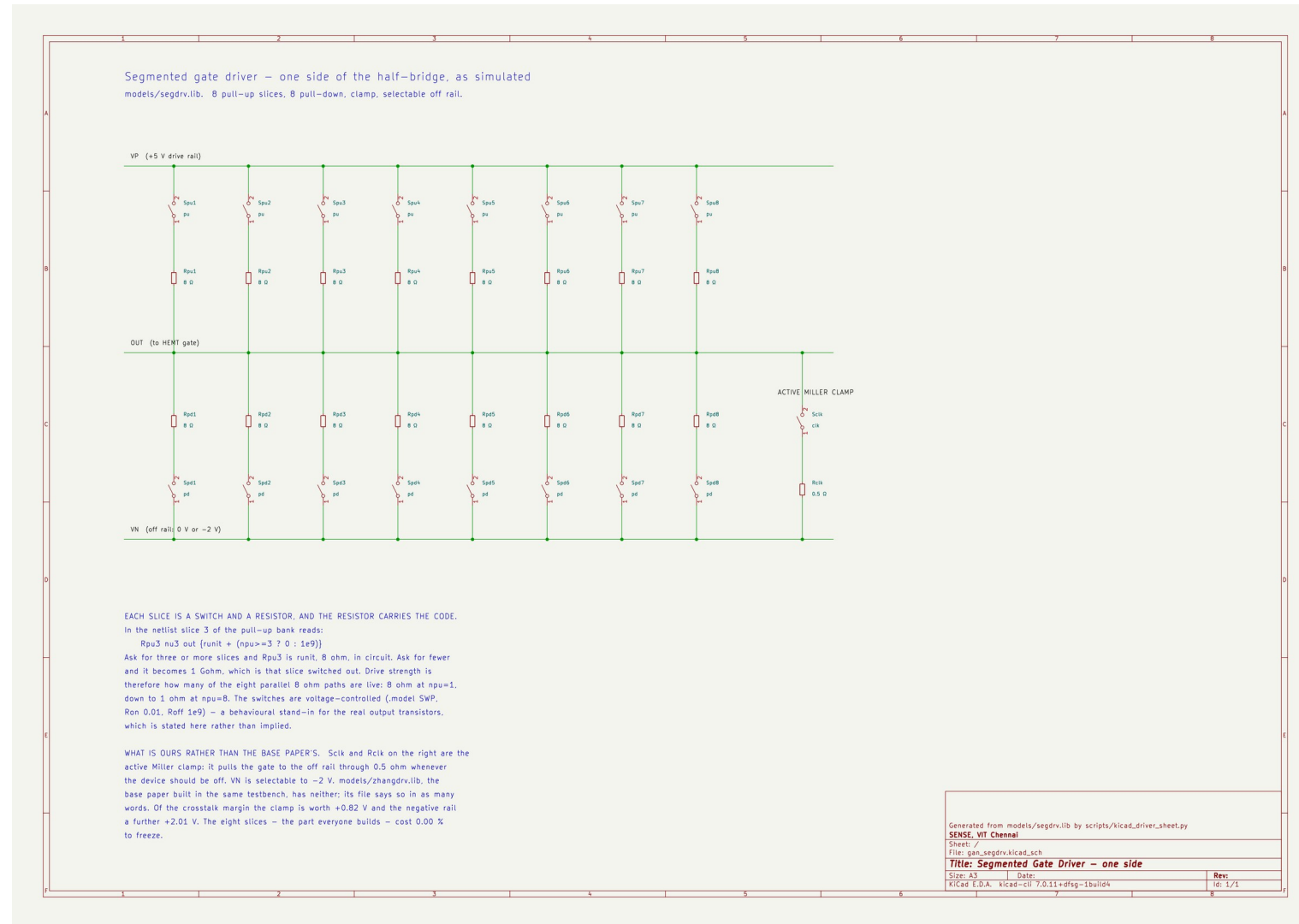


Fig. 6 - [kicad/gan_segdrv.kicad_sch](#). Eight pull-up segments, eight pull-down, the active Miller clamp on the right, and the off rail selectable to -2 V. Each segment is one switch and one resistor; the resistor carries the control word.

Implementation: Base Paper Gate Driver

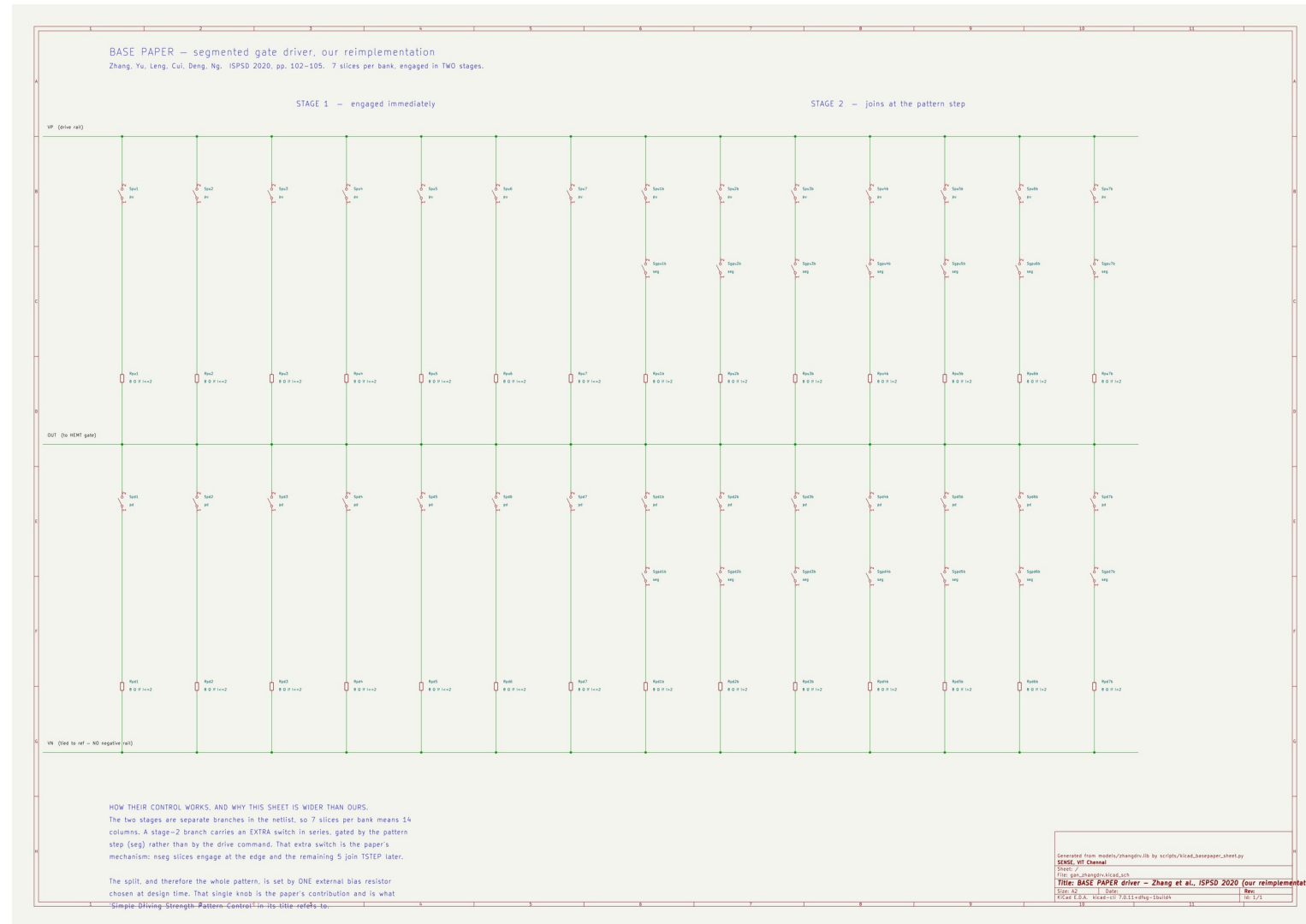


Fig. 7 - [kicad/gan_zhangdrv.kicad_sch](#), the base paper's driver reimplemented in the same testbench. Seven segments per bank in two stages, one bias resistor, no clamp branch, and the off rail tied to reference.

Segmented Driver: SPICE Implementation

models/segdrv.lib - the output stage varied throughout the study.

models/segdrv.lib

the segmented output stage under study

```
23 .subckt SEGDRV pu pd clk out vp vn ref params: npu=8 npd=8 runit=8 rclamp=0.5  ◀ npu / npd are the 4 + 4 bits the FPGA emits
. . .
26 Spu1 vp nu1 pu ref SWP
27 Rpu1 nu1 out {runit + (npu>=1 ? 0 : 1e9)}  ◀ in circuit when npu ≥ 1, and 1 GΩ out of it when not
28 Spu2 vp nu2 pu ref SWP
. . .
43 * ---- pull-down bank : slice enabled if npd >= i -----
44 Spd1 nd1 vn pd ref SWP
45 Rpd1 out nd1 {runit + (npd>=1 ? 0 : 1e9)}
. . .
61 * ---- active Miller clamp : strong, separately timed -----
62 Sclk out nclk clk ref SWP  ◀ the clamp: its own switch, its own timing
63 Rclk nclk vn {rclamp}  ◀ 0.5 Ω to the off rail — where the injected charge goes
. . .
65 .ends SEGDRV
```

Fig. 8 - Eight pull-up segments and eight pull-down, each a switch and a resistor. A segment is in circuit when the control word reaches it and 1 GΩ out of it when it does not, which is how 4 + 4 bits from the FPGA become a drive strength. The clamp is not one of the segments: it has its own switch, its own timing and its own 0.5 Ω path to the off rail, because it has to hold the gate down while the other device switches. The segments are discrete by design: a single variable resistor would simulate identically and could not be laid out.

Methodology

Every comparison is the same file with one thing changed. The following is held identical; the listed parameter is the only one that varies.

HELD IDENTICAL

the netlist [sim/dpt.cir](#), byte for byte - not a copy, not a variant

the device [models/egan.lib](#) on both sides, same V_{th} , same $R_{ds(on)}$, same C_{gs}

the parasitics 3 nH power loop, 0.3 Ω loop resistance, same gate inductance

the operating point 100 V bus, 10 A load, 25 °C junction

the solver same timestep, same options, same initial conditions

ENVIRONMENT ngspice-42 and KiCad 7.0.11 on a Debian container, headless, no hardware in the loop. Both are open source; there is no licensed tool and no vendor model anywhere in this project.

WHAT CHANGES

theirs vs ours the driver subcircuit, and nothing else:
[models/zhangdrv.lib](#) ↔ [models/segdrv.lib](#)

GaN vs silicon the device model only, $R_{ds(on)}$ matched at the 25 m Ω class

clamp on/off one parameter, CLKEN; off rail, one parameter, VNEG

Fig. 9 - Their driver is additionally given a freedom its own paper does not have: [scripts/headtohead.py](#) searches both of its controls at every corner and it is run at the setting that search finds optimal for it, read out of [results/headtohead.txt](#). Ours runs one fixed control word at every corner. Any remaining difference is the driver. [results/RESULTS-SUMMARY.txt](#) names the script behind each number, and [review/check_consistency.py](#) fails the build if a slide and the data disagree.

Model Validation: Three Device Models

The same three configurations, run on three independent device models.

Crosstalk margin under three device models (V_{th} – peak V_{gs} on the OFF device; > 0 is safe)

	no clamp, 0 V off rail	clamp on, 0 V off rail	clamp + –2 V rail (shipped)
Behavioural, diode junction C models/egan.lib	-0.249 V	+0.570 V	+2.576 V
SKY130 transistors scripts/silicon_check.py	-0.563 V	+0.031 V	+2.032 V
Charge-based junction C models/egan_c.lib	+0.115 V	+0.800 V	+2.710 V

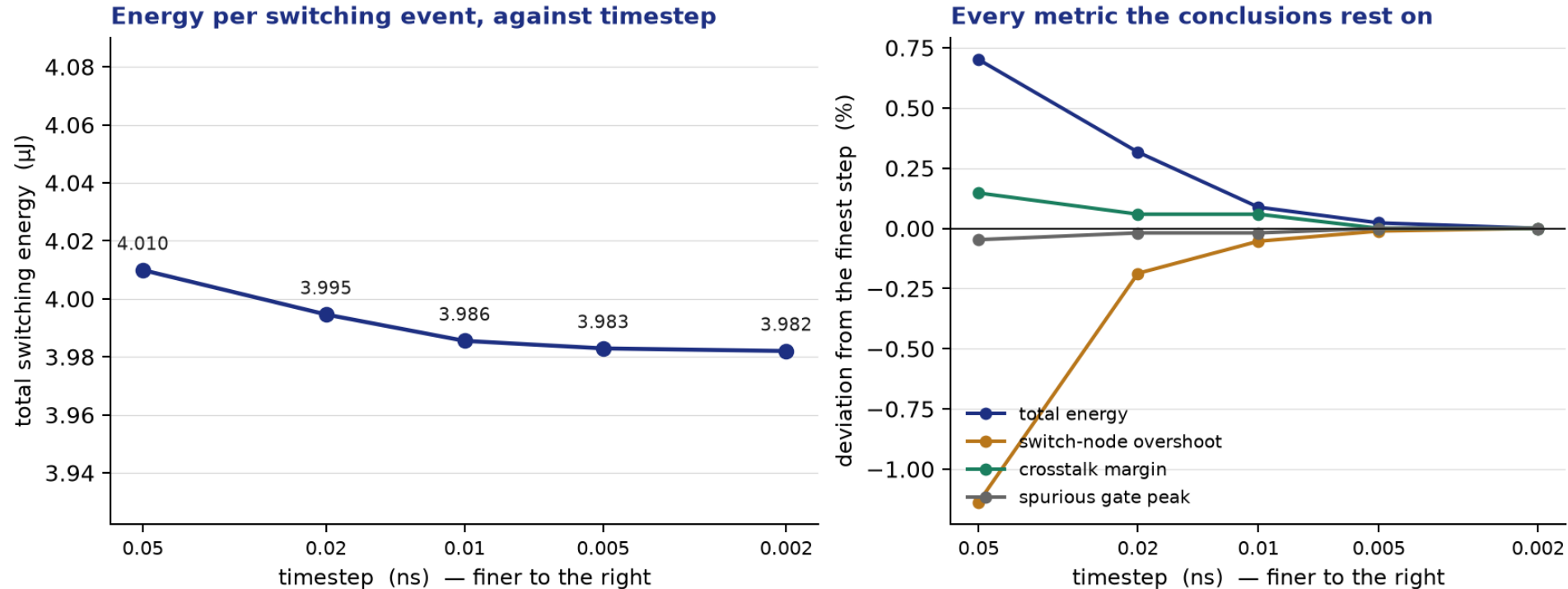
The SIGN of the no-clamp margin is model-dependent: on the charge-based model the OFF gate does not reach threshold, so on that model the fault does not occur.

The shipped configuration is positive on all three, +2.032 to +2.710 V. The ordering of the three configurations is identical under every model.

Fig. 10 - Simulated in ngspice. [models/egan.lib](#) is behavioural with diode junction capacitances; [scripts/silicon_check.py](#) runs real SKY130 transistors; [models/egan_c.lib](#) replaces the diodes with a charge-based C(V). The sign of the no-clamp margin depends on the model: on the charge-based model the OFF gate reaches only +0.115 V of margin, meaning it never crosses threshold and the fault does not occur, against –0.249 V behaviourally and –0.563 V on SKY130. What every model agrees on is the ordering of the three configurations, and that the shipped configuration is safe on all of them by +2.032 to +2.710 V. The claim this project makes is the second one.

Numerical Reliability

Whether these numbers are physics or solver settings, and every run accounted for.



scripts/metric_converge.py, five timesteps over a 25x range. Total energy moves 0.70 %, switch-node overshoot 1.14 %, the crosstalk margin 0.14 % and the spurious gate peak 0.04 %. Flat is the result.

Fig. 11 - [scripts/metric_converge.py](#) re-runs one word at five timesteps spanning 25x. The crosstalk margin the result rests on moves 0.14 % over that range and the spurious gate peak 0.04 %; [scripts/verdict_stability.py](#) flips 0 of 80 feasibility verdicts. Separately, the 36-corner grid is 25,911 completed runs of 25,920. [scripts/failed_runs.py](#) re-runs the nine that are not there: all nine fail again, so they are reproducible rather than flaky, all nine abort with the same ngspice transient convergence failure at the high-side gate node, and all nine are half-fixed settings - clamp without the rail, or rail without the clamp. None is the shipped configuration. One figure appears twice at different resolutions and the deck quotes the finer: switch-node overshoot is 17.9 % at the 0.02 ns step this deck reports, against 15.1 % at the sweep's 0.2 ns step, which is what [proof/LIVE-BUCK.sh](#) prints and says.

Base Paper Driver: Simulation Result

[models/zhangdrv.lib](#) on [sim/dpt.cir](#). Seven segments per bank in two stages, no clamp branch, no negative off rail.

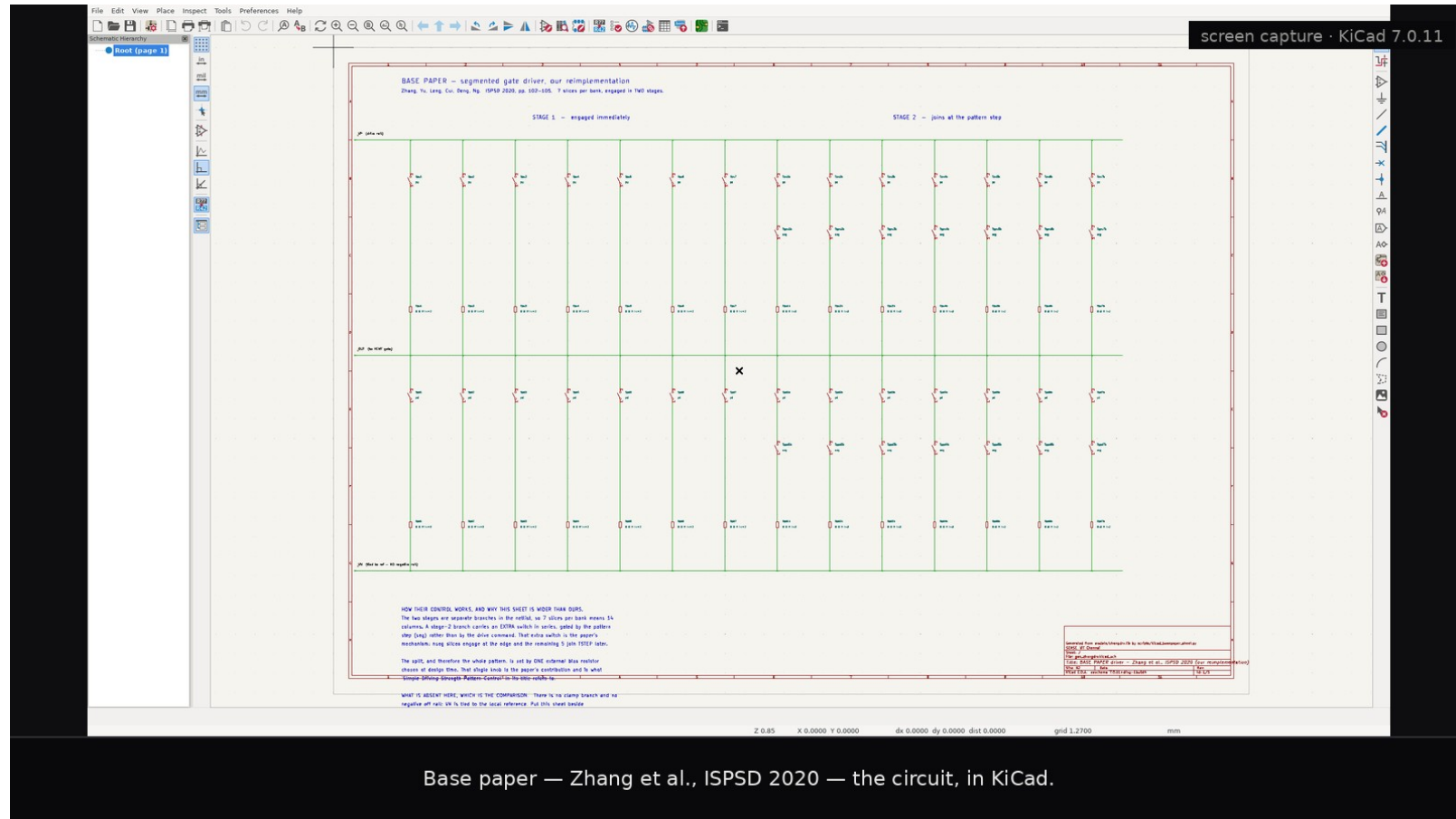
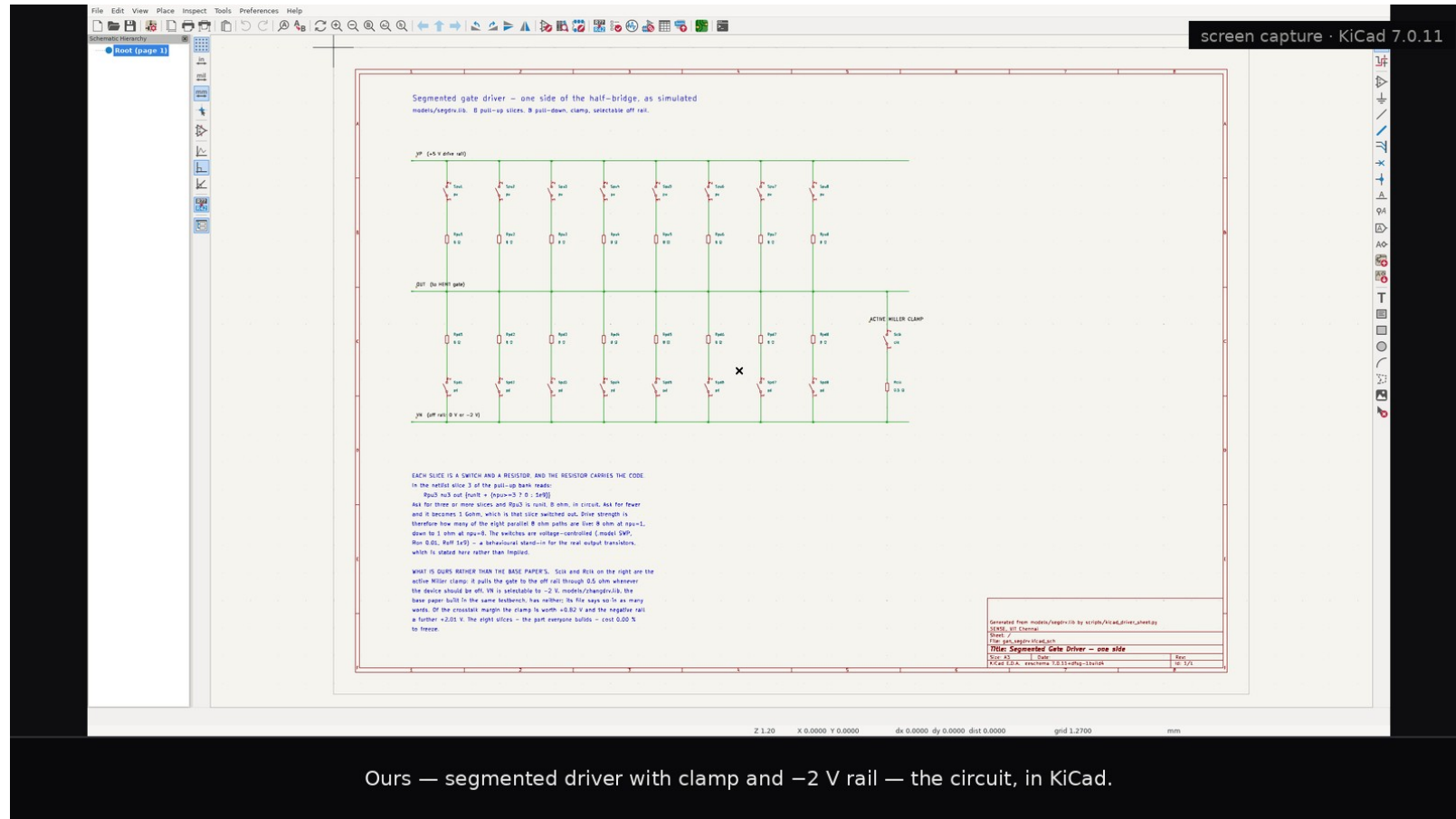


Fig. 12 - Their circuit, ngspice running it, the waveform. OFF gate peaks +0.993 V against a 1.400 V threshold - 0.407 V of margin. Run at nseg=2, tstep=4n, the setting [scripts/headtohead.py](#) found best for their driver here. 35 s.

Proposed Driver: Simulation Result

[models/segdrv.lib](#) on the same [sim/dpt.cir](#). Eight segments per bank, active Miller clamp, -2 V off rail.



Ours — segmented driver with clamp and -2 V rail — the circuit, in KiCad.

Fig. 13 - The same sequence, our driver. Gate peaks -1.176 V - 2.576 V of margin, $6.3\times$ our implementation of their scheme. One fixed control word. Axes identical to the previous slide. 35 s.

Live Simulation

bash [proof/LIVE-SIM.sh](#) - two ngspice transients, about ten seconds, on the presentation machine.

```
$ bash proof/LIVE-SIM.sh 5.0 s

GaN half-bridge -- crosstalk, simulated live
-----
machine  vm      2026-09-27 13:24:17
spice    ** ngspice-42 : Circuit level simulation program
circuit  sim/dpt.cir      device  models/egan.lib
changing ONLY  CLKEN (clamp) and  VNEG (off rail)

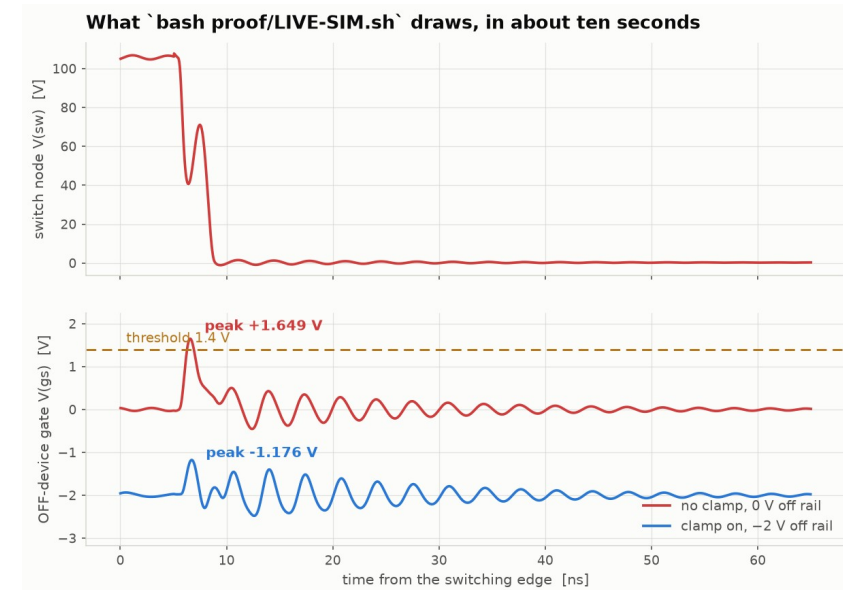
(a) fastest drive, no clamp, 0 V off rail ...
    OFF-device gate reaches +1.6488 V  against a 1.400 V threshold
    margin -0.2488 V    false_turn_on = 1  SHOOT-THROUGH

(b) same circuit, clamp on, -2 V off rail ...
    OFF-device gate reaches -1.1759 V
    margin +2.5759 V    false_turn_on = 0  SAFE

-----
                                on the slide    this run
OFF gate, no clamp [V]          1.65           1.6488
margin, shipped [V]             2.576          2.5759
-----
two ngspice transients, 3.7 s wall clock on this machine

drawing the waveforms those two runs produced ...
```

recorded run of the same command



the waveforms it produces

Fig. 14 - Simulated in ngspice. Two runs of one circuit file. Between them only CLKEN (the Miller clamp) and VNEG (the gate off rail) change, so no other parameter varies. It prints what it measured beside what these slides claim - OFF gate 1.649 V against the deck's 1.65, margin 2.576 V against the deck's 2.576 - then draws the waveforms those two runs produced. 3.8 s of that is ngspice; the rest is drawing. [scripts/live_demo.py](#).

Crosstalk: Fault and Mitigation

Crosstalk at low-side turn-on, 100 V / 10 A — models/egan.lib

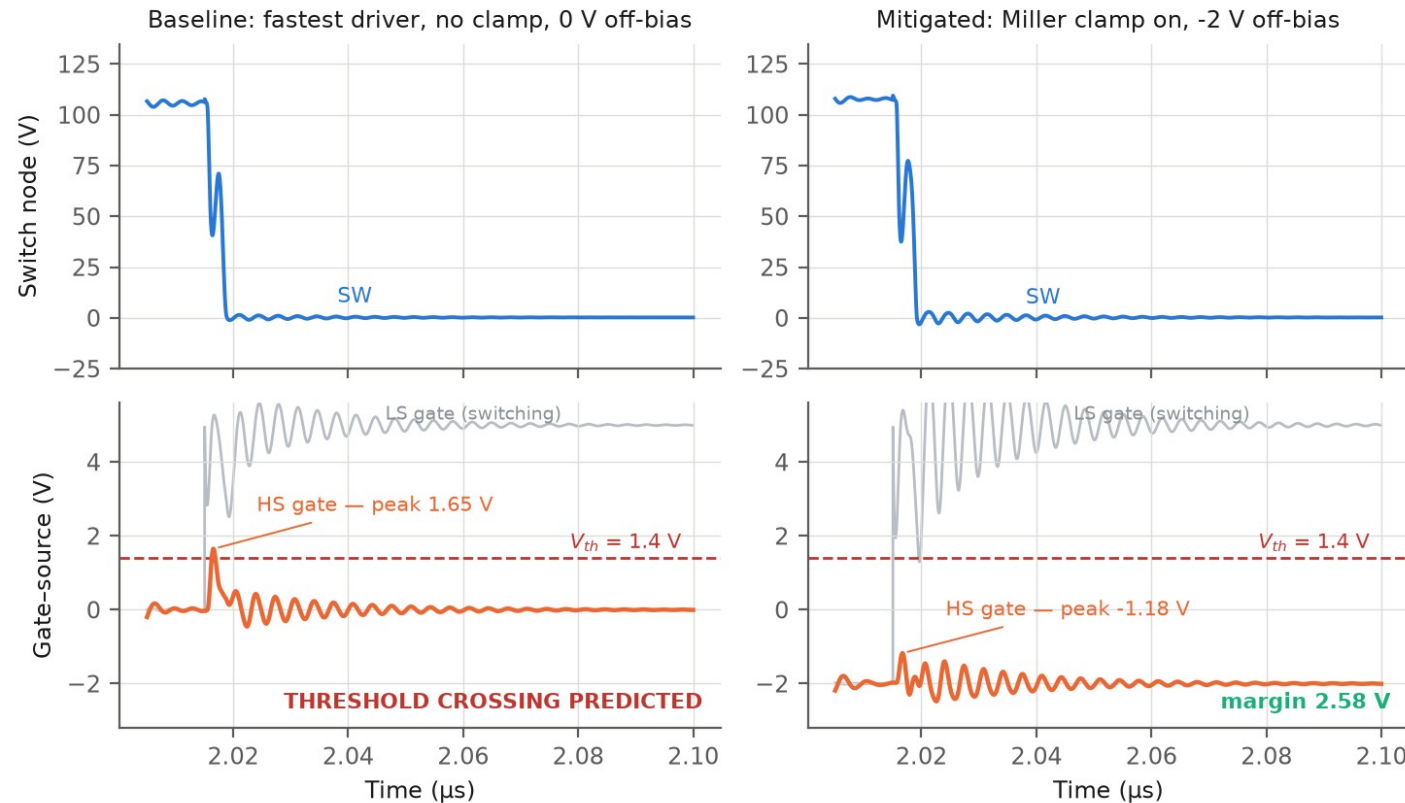


Fig. 15 - Simulated in ngspice. Two runs of [sim/dpt.cir](#). Top row: the switch node falls from 106 V, its 90-10 % edge taking 2.90 ns and peaking at 124 V/ns - that dv/dt drives $i = C_{GD} \cdot dv/dt$ into the OFF device's gate, and is the cause. Bottom row: that gate. Left, without the clamp, it rests at +0.005 V, is lifted +1.643 V and peaks at +1.649 V, past the 1.4 V threshold - `false_turn_on` = 1. Right, with the clamp and the -2 V rail, it rests at -1.941 V, is lifted only +0.765 V and peaks at -1.176 V - `false_turn_on` = 0. The two fixes are not one mechanism: the rail moves where the gate starts, the clamp shortens the lift by giving the injected charge a 0.5 Ω path out. [scripts/waveform_anatomy.py](#).

Comparison with the Base Paper

Only the driver changes. Ours holds one fixed word at every corner; theirs is re-optimised at each one - more freedom than their design has. At three of four corners that search returns their built setting, so those two columns match.

crosstalk margin	base, as their paper builds it	base, re-tuned at every corner	ours, one fixed word	vs their best
50 V / 2 A / 25 °C	+0.338 V	+0.503 V	+2.757 V	5.5×
100 V / 10 A / 25 °C	+0.407 V	+0.407 V	+2.576 V	6.3×
200 V / 2 A / 125 °C	+0.261 V	+0.261 V	+2.309 V	8.9×
200 V / 10 A / 125 °C	+0.181 V	+0.181 V	+2.251 V	12.4×

The margin gap widens with stress

Their margin falls +0.503 → +0.181 V from the mildest corner to the hottest, at their best setting for each. Ours goes +2.757 → +2.251 V. They degrade where it matters most; a clamp does not care how hot the device is.

Worst corner is what a converter has to survive: +0.181 V against +2.251 V.

The edge is faster, not slower

Their scheme reduces crosstalk by slowing the edge: 67–103 V/ns at the switch node. Ours runs 101–175 V/ns - about twice as fast - and still wins by a factor of several.

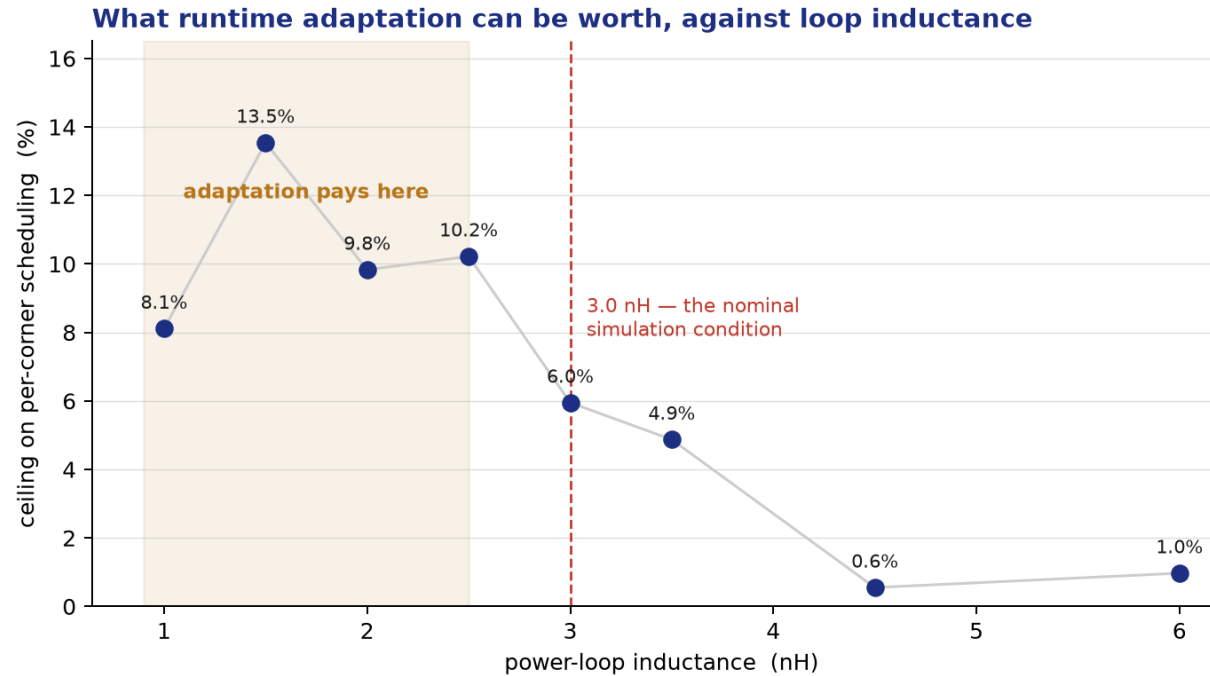
The margin is not bought with switching speed. That is the useful form of the result.

What it costs. Turn-on energy higher at 3 of 4 corners. Their driver: one bias resistor. Ours: a clamp device, a negative supply, 20 LUTs.

What we do not claim. Their netlist is unpublished. Every ratio here is against our implementation of their scheme, not their measured result.

When Runtime Adaptation Is Worth Building

The headline is conditional, and this is the condition.



Eight sweeps, `scripts/loop_sweep.py` → `loop_analyse.py`. Points, not a fitted curve: the series peaks at 1.5 nH and is non-monotonic below 2.5 nH, so a smooth trend would assert more than eight runs support. Loop inductance is set by board layout, not by the transistor.

Fig. 16 - Simulated in ngspice, [scripts/loop_sweep.py](#) into `loop_analyse.py`. The ceiling on per-corner scheduling is 13.5 % at 1.5 nH and 0.55 % at 4.5 nH, so the answer to "is adaptive gate control worth the sensor, the lookup table and the controller" is not a property of the driver: it is a property of the board the driver sits on. Below about 2.5 nH the answer is yes and above it the answer is no. The series is not monotonic in that band, which is why it is drawn as eight points rather than as a curve. 3.0 nH is our nominal simulation condition, not a measured layout, and the 2.6 % headline is the value at that nominal condition.

The Cost of the Fix

What the crosstalk margin is paid for with.

What the crosstalk fix buys, and what it costs

	without the fix	shipped configuration	change
crosstalk margin, OFF device	−0.249 V	+2.576 V	+2.825 V
converter efficiency	97.73 %	97.49 %	−0.24 points
converter loss	5.502 W	6.071 W	+0.570 W
switch-node peak, 100 V bus	106 V	118 V	+12 V
turn-on energy, 100 V / 10 A	3.883 μJ	4.035 μJ	+0.152 μJ

The fix is a trade, not a free win: it buys +2.825 V of gate margin for 0.24 points of efficiency and 12 V of extra switch-node stress — 59 % of a 200 V device rating, so stress is not the binding constraint at this bus.

Fig. 17 - Simulated in ngspice, from [results/buck_sweep.csv](#), [results/panel_metrics.csv](#) and [results/headtohead.csv](#). The fix buys +2.825 V of gate margin, and it is bought with 0.24 points of converter efficiency, 0.570 W of loss and 12 V of extra switch-node peak. About 0.22 W of that loss is third-quadrant conduction: GaN has no body diode, so during dead time the device conducts in reverse at $V_{th} + |V_{off}| + I \cdot R_{ds(on)}$, and a deeper off rail makes that drop larger. At 118 V on a 200 V-rated device the stress is 59 % of rating, so it is a real cost but not the binding constraint at this bus.

Conclusions

- 1. Gate robustness.** An always-on Miller clamp with a -2 V off rail raises the simulated off-state gate margin on all three device models tried, from -0.249 V to $+2.576$ V on the behavioural model and to between $+2.0$ and $+2.7$ V on the other two.
- 2. Adaptation.** Choosing one good fixed control word is worth 26.5 % of the baseline. Re-tuning it while the converter runs adds 2.6 % more - 8.9 % of the total gain - at our nominal 3 nH power loop.
- 3. That answer is conditional.** The ceiling on scheduling runs from 13.5 % at 1.5 nH to 0.55 % at 4.5 nH. Whether adaptive gate control earns its hardware is decided by board layout, not by the driver.
- 4. It is a trade.** The margin costs 0.24 efficiency points, 0.570 W and 12 V of switch-node stress, and raises turn-on energy at three of four corners.
- 5. Status.** Simulation only. No hardware built. The comparison is against our own implementation of the published baseline, because the authors' netlist is not published.
- 6. What would change this.** A measured gate waveform that does not match the simulated one, or an extracted loop inductance below 2.5 nH - which we have not measured, and which would invert conclusion 3.

Next Steps

What remains is bench work, not further simulation.

Place-and-route on a chosen board. Synthesis is done; the flow stops there because the XDC pins are placeholders. Doing it properly also means driving the 200 MHz clock from an MMCM rather than straight off a pin, which is what makes 34 clock-to-pin paths fail today.

A hardware half-bridge, measured. This is the whole of the remaining risk. Everything in this deck is a simulation of a converter that has never been built, and one behavioural GaN model underlies all of it.

Additionally: transcribe the silicon MOSFET datasheet digits rather than using datasheet-class values, and re-run the ceiling on the transistor-level stage now that it is known to work.

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The papers this design engages with directly. [10] is the base paper, reimplemented in [models/zhangdrv.lib](#) and measured against throughout.

Thank you

GaN Based Synchronous Buck Converter with an Improved Gate Driver

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Everything in this deck runs on request: github.com/Amritha902/gan-driver